

Final Project Report

TransLingua: AI-Powered Multi-Language Translator

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1. INTRODUCTION

1.1 Project Overview

TransLingua is an AI-Powered Multi-Language Translation application developed using **Streamlit** and **Google Gemini API**. The system is designed to simplify and automate real-time language translation by accurately translating text between multiple languages with intelligent language detection. In today's global digital environment, communication across different languages has become essential. However, language barriers create challenges in education, business communication, travel, research, and online interactions. Traditional translation tools often require manual language selection, may not detect context properly, and sometimes fail to produce natural-sounding translations. The proposed system addresses these challenges by integrating **Generative AI and Large Language Models (LLMs)** to deliver context-aware and accurate translations. Users simply enter text into the application, and the system automatically detects the source language and translates it into the desired target language instantly.

The system allows users to:

1. Select source and target languages dynamically
2. Automatically detect input language
3. Swap languages easily using a swap button
4. Generate fast and structured translations
5. Support multiple global languages

Built using **Streamlit** for the interactive frontend interface and **Google Gemini API** for AI-powered language processing, the application ensures:

- Real-time multi-language translation
- Automatic language identification
- Context-aware and natural translations
- User-friendly interface
- Reduced communication barriers

The system can be used by students, professionals, travelers, researchers, and organizations requiring quick and reliable language translation support. Overall, TransLingua demonstrates the practical implementation of Generative AI in solving real-world communication problems by improving accessibility, enhancing global interaction, and enabling seamless cross-language communication.

1.2 Purpose

The primary purpose of TransLingua is to eliminate language barriers by providing an intelligent AI-based translation system that delivers fast, accurate, and context-aware translations across multiple languages. Traditional translation methods often require users to manually choose source languages and rely on basic rule-based translation systems, which may not preserve meaning, tone, or contextual accuracy. This creates inefficiencies and misunderstandings, particularly in professional, academic, or international communication.

The purpose of this project is to design and implement a system that:

1. Automatically detects the source language
2. Translates text into the selected target language
3. Provides context-aware AI-generated translations
4. Reduces manual effort in cross-language communication
5. Enhances global accessibility

The system leverages **Google Gemini API (LLM)** for intelligent language understanding and translation generation, while **Streamlit** provides an interactive and responsive web interface for users. Additionally, this project demonstrates the real-world application of Generative AI and Large Language Models in the field of Natural Language Processing (NLP). It showcases how AI can:

- 1) Improve communication efficiency
- 2) Support multilingual collaboration
- 3) Enable global connectivity
- 4) Increase productivity

In summary, the purpose of this project is to:

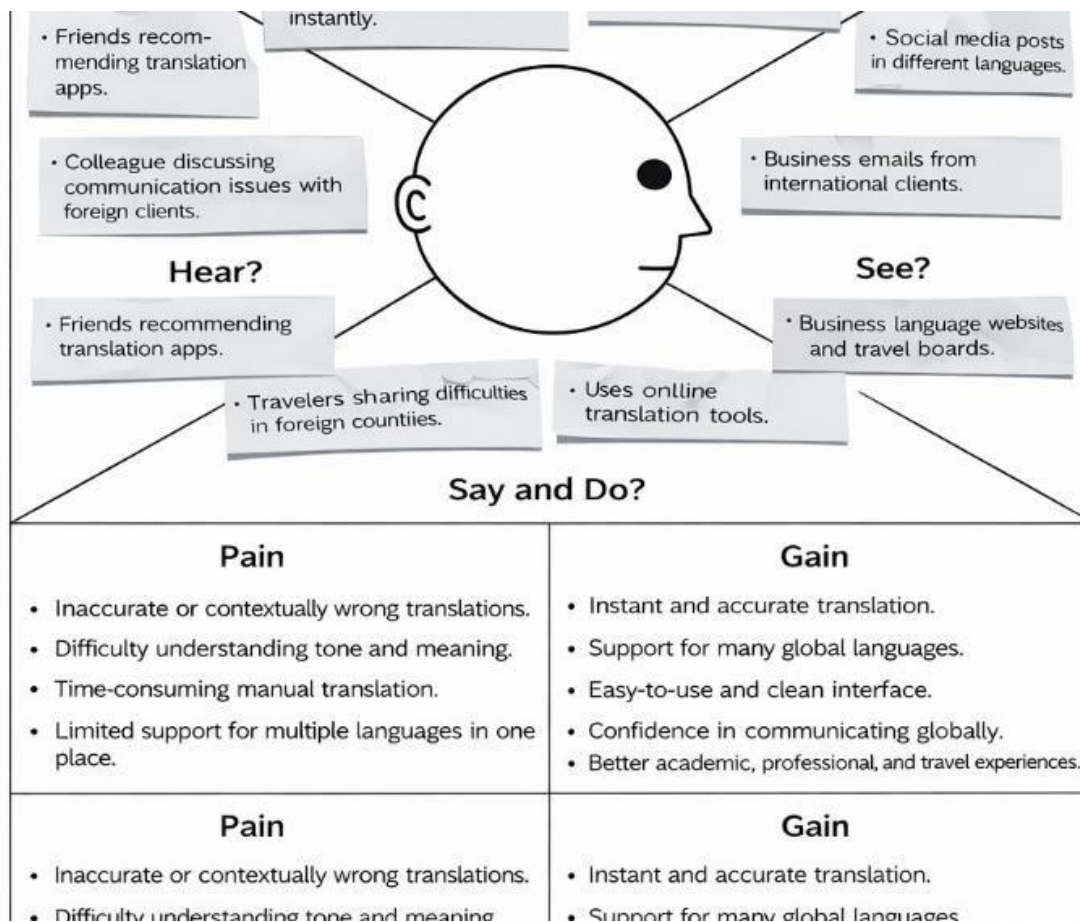
- Automate multilingual text translation
- Provide intelligent and context-aware outputs
- Minimize communication barriers
- Enhance user experience through AI integration
- Demonstrate practical implementation of Generative AI in language processing

2. IDEATION PHASE

2.1 Problem Statement

Language barriers continue to create significant communication challenges in today's globalized world. Individuals frequently struggle to understand and communicate across different languages in education, business, travel, research, and online interactions. Miscommunication can lead to confusion, reduced productivity, missed opportunities, and inefficient collaboration. Many existing translation tools require users to manually select source languages and often provide literal translations without understanding context, tone, or intent. This can result in inaccurate or unnatural outputs, especially when translating complex sentences, idioms, or technical content. Additionally, switching between multiple translation platforms for different languages can be inconvenient and time-consuming. Similarly, professionals such as students, researchers, multinational employees, and content creators require fast, reliable, and context-aware translations to communicate effectively. Manual translation processes are repetitive, inefficient, and prone to errors. There is a clear need for an intelligent system that can automatically detect the input language and generate accurate, context-aware translations in real time. Such a solution should reduce communication barriers, minimize manual effort, and provide structured, natural-sounding translations across multiple global languages. TransLingua addresses this problem by leveraging Generative AI through Google Gemini API to perform automatic language detection and AI-powered multilingual translation. Built with an interactive Streamlit interface, the system ensures faster, smarter, and more efficient cross-language communication for individuals and professionals alike.

2.2 Empathy Map Canvas



2.3 Brainstorming

Step-1: Team Gathering, Collaboration and Select the Problem Statement

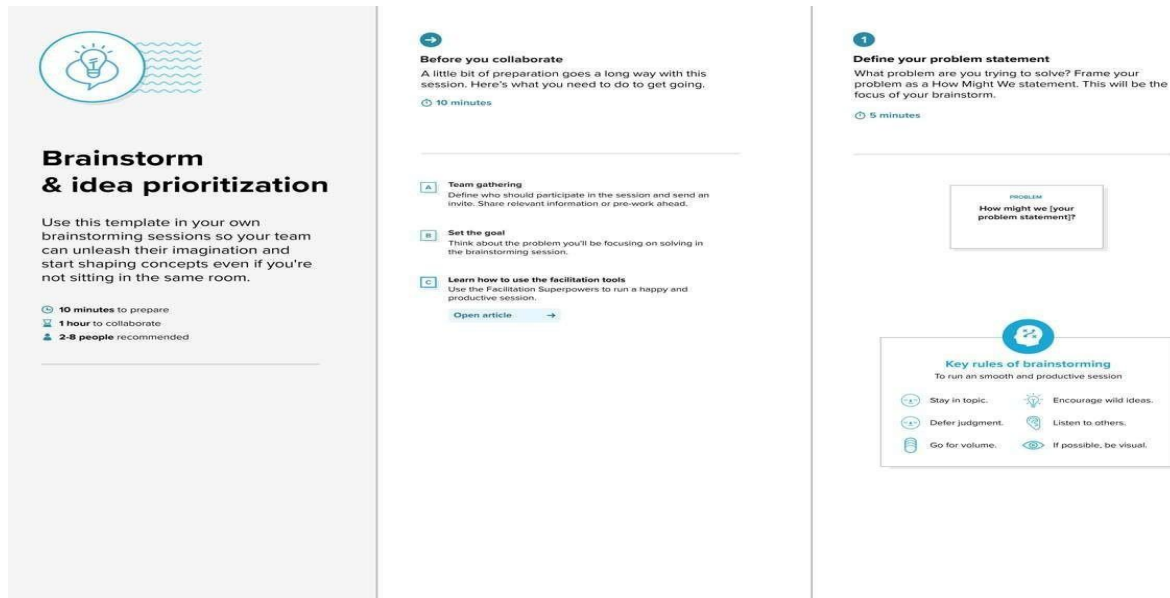
In the initial phase of the TransLingua project, the team comes together to align on goals, responsibilities, and expectations. The focus is on understanding the communication challenges faced by students, professionals, travelers, and global businesses due to language barriers.

Key activities:

- Built a collaborative environment for sharing ideas and understanding real-world language challenges.
- Conducted brainstorming sessions to explore pain points such as: Difficulty in communicating across different languages, Lack of instant and accurate translations, Limited access to multilanguage tools, Time-consuming manual translation processes.
- Identified a relevant, meaningful, and user-centered problem statement.
- For example: "Individuals and organizations need a fast, accurate, and intelligent way to translate text across multiple languages without relying on complex tools or manual effort."

Outcome:

A clear problem statement that guides the design and development of TransLingua, ensuring the team is aligned and focused on solving real communication challenges across languages



Step-2: Brainstorm, Idea Listing and Grouping

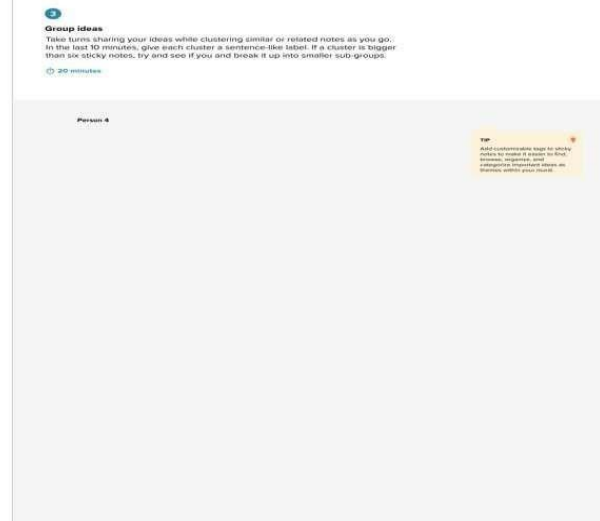
Once the problem statement is defined, the team moves into ideation to explore potential solutions for Explore with TransLingua.

Key activities:

- Brainstormed a wide range of ideas for the application, including features such as:
 - a) AI-powered real-time text translation
 - b) Automatic language detection
 - c) Support for multiple global languages
 - d) Swap language functionality
 - e) Clean and simple user interface
 - f) Integration with advanced AI models (Gemini API)
- Encouraged all ideas, no matter how small or innovative.
- Documented and grouped ideas in a shared workspace to visualize feature clusters and technical feasibility.

Outcome: A comprehensive list of potential features and functionalities for TransLingua, categorized into themes such as:

- AI Translation Engine
- User Experience Enhancements
- Multi-Language Support
- Automation & Smart Detection



Step-3: Idea Prioritization

After gathering and grouping ideas, the team prioritizes features and solutions to focus on the highest value and most feasible elements for TransLingua.

Key activities:

- Evaluated ideas based on:
 - User impact
 - Technical feasibility
 - Development effort
- Used prioritization techniques such as:
 - MoSCoW Method (Must Have, Should Have, Could Have, Won't Have)
 - Impact vs. Effort Matrix
 - Team consensus
- Decided feature implementation order:
- Must-Have:
 - ◆ AI-powered text translation
 - ◆ Support for multiple languages
 - ◆ Automatic language detection
- Should-Have:
 - ◆ Swap language button

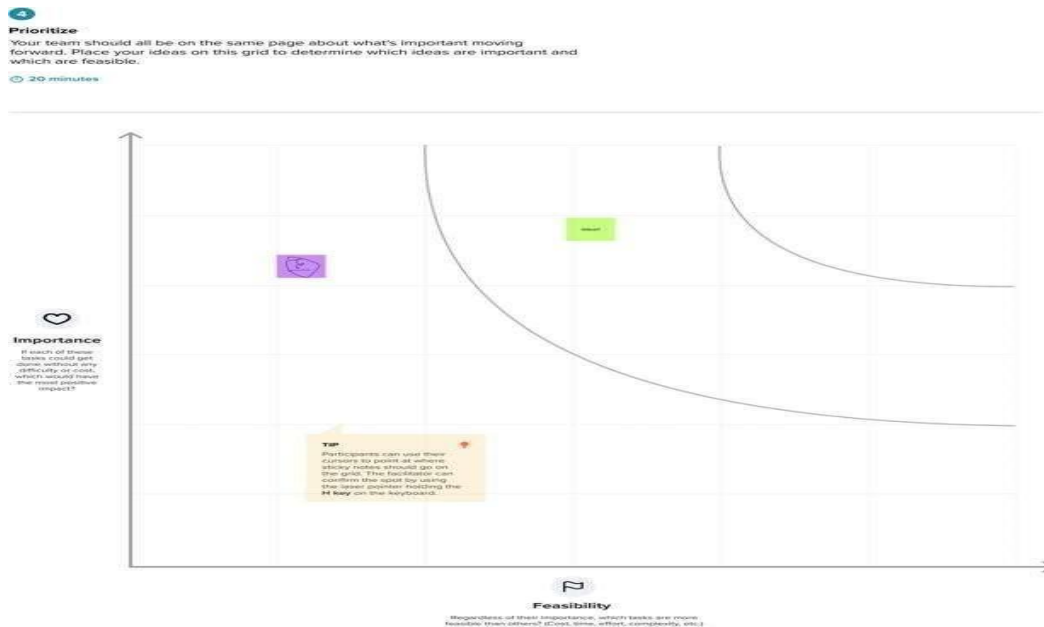
- ◆ Dynamic language dropdown selection

- Could-Have:

- ◆ Download/export translated text

- ◆ Voice input/output support

Outcome: A focused development roadmap for TransLingua that maximizes user value while remaining realistic within the project timeline. This ensures that core translation functionality is implemented first without overcomplicating the system.



3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

The customer journey for **TransLingua** describes the steps a user takes from discovering the application to receiving accurate multilingual translations, highlighting their actions, thoughts, and experiences throughout the process.

1. Awareness

Users first learn about TransLingua through academic projects, online demonstrations, social media, AI-related discussions, or recommendations from peers. At this stage, users may feel frustrated due to language barriers in communication, study materials, or professional tasks. The idea of an AI-powered multi-language translator creates curiosity and interest.

2. Interest

After discovering the platform, users open the Streamlit web application to explore its features. They look for capabilities such as automatic language detection, fast translation, and support for multiple languages. Users may wonder whether the system provides accurate, context-aware translations rather than simple word-to-word conversion.

3. Input / Text Entry

Users enter or paste text into the input field. They may also select a target language (if required). At this stage, users expect the system to correctly understand the input language automatically. The application validates the input and prepares it for AI processing using the Gemini API.

4. AI Processing

Once the user submits the text, the Gemini Generative AI model processes the content. The system performs:

- Automatic language detection

- Context-aware translation

- Natural sentence restructuring

Users experience anticipation, expecting accurate and fluent output.

5. Output / Translation Display

The translated text is displayed in a clear, structured format within the Streamlit interface. Users feel satisfied if the translation is natural, accurate, and easy to understand. The system reduces the effort required to manually translate or switch between multiple tools.

6. Optional Actions

Users can:

- Copy the translated text

- Download the output

- Translate into another language

- Modify the input for refinement

These features improve usability and flexibility.

7. Feedback / Iteration

Users may adjust the input text or change the target language to generate improved results. This iterative process enhances personalization and ensures better translation accuracy.

3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

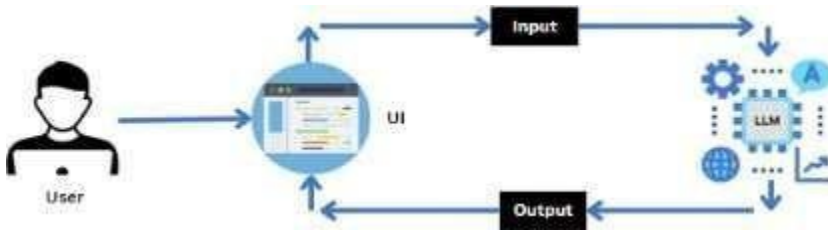
FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Input Interface	Enter or paste text for translation Optional: Select target language
FR-2	Automatic Language Detection	Detect source language automatically
FR-3	AI Translation Generation	Generate accurate, context-aware translated text Support multiple global languages
FR-4	Output Display	Display translated text clearly in UI
FR-5	Export & Copy Options	Copy translated text to clipboard Download translated output (optional)
FR-6	Error Handling	Validate empty input Handle API errors gracefully
FR-7	Multi-language Support	Allow translation between various supported languages

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system must provide a simple and intuitive interface for all users
NFR-2	Security	API calls to Gemini must be secured; user inputs must not be stored without consent
NFR-3	Reliability	The system must consistently generate accurate translations without failures
NFR-4	Performance	Translation results should be generated within 3–5 seconds
NFR-5	Availability	The application should maintain high uptime (≥ 99%)
NFR-6	Scalability	The system should support multiple users simultaneously
NFR-7	Extensibility	The architecture should allow future features like voice translation, document translation, or chat-based interaction

3.3 Data Flow Diagram



A **Data Flow Diagram (DFD)** represents the flow of data within the system, showing how information moves from inputs to processes and finally to outputs. For **TransLingua**, the DFD illustrates how user inputs are processed by the system to generate personalized travel itineraries using Streamlit and the PaLM Generative AI model.

3.4 Technology Stack



The technology stack for this project has been carefully selected to build a lightweight, interactive, and AI-driven multi-language translation application. The system leverages **Streamlit** for the user interface and **Google Gemini API (Generative AI)** for intelligent language detection and translation.

1. Frontend / User Interface

Technology: Streamlit

- Streamlit is a Python-based framework used for building interactive web applications quickly and efficiently.
- It is used to design a clean and user-friendly interface where users can:
 - Enter or paste text for translation
 - Select target language
 - Use automatic language detection
 - Swap source and target languages
- Displays translated text in a structured and readable format.

- Provides interactive components such as:
 - Text input fields
 - Dropdown menus
 - Buttons
 - Dynamic output display

Streamlit ensures a responsive and easy-to-use interface without requiring complex frontend development.

2. Backend / Application Logic

Technology: Python

- Handles processing of user inputs.
- Validates input text before translation.
- Constructs structured prompts for the AI model.
- Manages secure communication with the Gemini API.
- Parses and formats the AI-generated translation.
- Connects frontend UI with the AI engine smoothly.

Python ensures efficient data handling and seamless integration between user inputs and AI-generated outputs.

3. Generative AI Engine

Technology: Google Gemini API (Large Language Model)

- Gemini is an advanced Large Language Model (LLM) capable of understanding and generating human-like text.
- Performs automatic source language detection.
- Generates context-aware and natural translations.
- Maintains sentence structure and meaning during translation.
- Supports multiple global languages dynamically.

The AI engine ensures intelligent translation rather than simple word-to-word conversion.

4. Deployment / Hosting

Technology: Streamlit Cloud / Local Deployment

- The application can be deployed on Streamlit Cloud for public access.
- Can also be run locally during development and testing.
- Cloud deployment ensures scalability and availability.
- Easily extendable to other cloud platforms (AWS, GCP, Azure) in the future.

4. PROJECT DESIGN

4.1 Problem Solution Fit

Language barriers continue to create significant communication challenges in education, business, travel, research, and digital interactions. Individuals such as students, professionals, travelers, and multinational employees often struggle to understand and communicate in different languages. Existing translation tools may require manual language selection, provide literal translations without contextual understanding, or fail to preserve tone and meaning.

These limitations lead to misunderstandings, reduced productivity, inefficient communication, and missed opportunities. Users frequently face issues such as:

- Inaccurate contextual translation
- Difficulty detecting source language
- Limited multi-language support
- Slow or inconsistent translation responses
- Dependence on multiple tools for different languages

Similarly, organizations and professionals dealing with international clients face repetitive and time-consuming manual translation processes. This affects communication efficiency and overall workflow productivity.

The core problem is the absence of a centralized, intelligent, and efficient AI-powered translation system that can automatically detect languages and provide accurate, context-aware translations instantly.

TransLingua Solution:

TransLingua addresses these communication challenges by providing an AI-powered web-based platform that performs intelligent multilingual translation using Generative AI.

Key features include:

- **Automatic Language Detection:** Identifies the input language without manual selection.
- **Context-Aware Translation:** Uses Google Gemini LLM to maintain sentence meaning and tone.
- **Multi-Language Support:** Supports translation between multiple global languages dynamically.
- **Real-Time Processing:** Provides fast and efficient translation output.
- **User-Friendly Interface:** Simple Streamlit-based interface for easy interaction.
- **Language Swap Functionality:** Allows quick switching between source and target languages.
- **Extensibility:** Future scope includes voice translation, document translation, and chat-based translation.

By automating intelligent multilingual communication, TransLingua reduces language barriers, improves productivity, and enhances global connectivity for individuals and organization

Purpose of Problem–Solution Fit

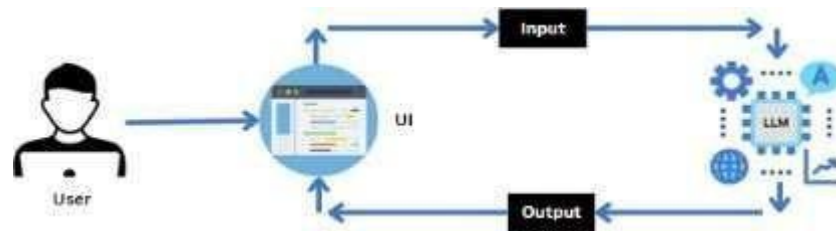
Defining the Problem–Solution Fit for TransLingua ensures the system addresses real and validated multilingual communication challenges:

- **Solve actual communication pain points:** Reduce language barriers and translation errors.
- **Improve accessibility:** Provide easy access to accurate multilingual translation.
- **Enhance productivity:** Minimize manual translation effort and repetitive tasks.
- **Increase reliability and trust:** Deliver context-aware AI-generated translations.
- **Deliver immediate value:** From the first interaction, users receive fast and accurate translations tailored to their needs.

4.2 Proposed Solution

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Many individuals, students, travelers, researchers, and businesses face communication barriers due to language differences. Existing translation tools often lack contextual understanding, provide inaccurate translations, or require complex installations. There is a need for an AI-powered, easy-to-use, web-based solution that provides fast and accurate multi-language translation.
2	Idea / Solution Description	TransLingua is a web-based AI-powered multi-language translator built using Streamlit and Google Gemini API. The system detects the source language automatically, translates text into the selected target language, and displays the output instantly. It provides a simple interface with language selection, swap functionality, and real-time translation capabilities.
3	Novelty / Uniqueness	• Automatic language detection feature • Uses advanced Generative AI (Gemini 2.5 model) for context-aware translation • Clean and lightweight web interface • No installation required (browser-based) • Dynamic language handling and swap option
4	Social Impact / Customer Satisfaction	The solution reduces communication gaps globally, supports students and researchers in academic work, assists travelers in foreign countries, and helps businesses expand internationally through seamless multilingual communication.
5	Business Model (Revenue Model)	• Freemium model (basic free, premium paid) • API usage subscription for enterprises • Advertisement-based model for free users • SaaS model for organizations requiring bulk translations
6	Scalability of the Solution	The system is scalable because it uses cloud-based AI APIs. The backend can handle multiple requests simultaneously. Deployment on cloud platforms such as Streamlit Cloud, AWS, or GCP enables horizontal scaling based on user demand.

4.3 Solution Architecture



1. Overview

TransLingua is a web-based AI-powered multi-language translation platform that allows users to enter or paste text and receive accurate, context-aware translations instantly. The system automatically detects the source language and translates it into the selected target language using Generative AI.

The platform is designed for simplicity, speed, accuracy, and scalability, ensuring seamless multilingual communication for students, professionals, researchers, and businesses.

2. High-Level Components

A. User Interface (Frontend)

Technology: Streamlit (Python)

Functionality:

- Input field for entering or pasting text
- Dropdown menu for selecting target language
- Optional automatic source language detection
- Swap language button
- Submit/Translate button
- Display of AI-generated translated text in a structured format
- User-friendly error messages for invalid inputs
- Option to copy or download translated output

B. API Layer / Backend

Technology: Python (Streamlit backend / Flask / FastAPI – if extended)

Functionality:

- Receives user input from frontend
- Validates text input
- Constructs structured prompt for AI model
- Sends request to Gemini API
- Handles API responses and errors
- Returns formatted translation output to frontend

C. AI Engine

Technology: Google Gemini API (Generative AI – LLM)

Functionality:

- Automatically detects source language
- Processes input text contextually
- Generates accurate, natural translations
- Preserves tone and sentence structure
- Supports translation across multiple global languages
- Ensures structured output suitable for UI rendering

D. Data Layer (Optional / Future Scope)

Technology: Cloud Database (MongoDB / PostgreSQL – Future Integration)

Functionality:

- Store translation history
- Store user preferences
- Maintain usage logs
- Track analytics for monitoring system performance

(Current implementation does not require persistent storage unless history feature is enabled.)

E. Export & Reporting Module

Functionality:

- Copy translated text to clipboard
- Download translation as text or PDF (future scope)
- Share translation via email or messaging platforms (future enhancement)

3. Key Features of Architecture

- Modular Design:** UI, backend logic, AI engine, and optional data layer are loosely coupled.
- Scalable:** Can support multiple users and concurrent AI requests.
- Extensible:** Future integration with voice translation, document translation, chatbot-based translation, or cloud deployment.
- Simple Workflow:** Input → AI Processing → Output, minimizing user complexity.
- Secure API Handling:** API keys managed using environment variables.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog, Sprint Schedule, and Estimation (TransLingua)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint 1	User Registration & Login	USN-1	As a user, I can register using email and password.	2	High
Sprint 1	User Registration & Login	USN-2	As a user, I can login securely using credentials.	1	High
Sprint 1	Language Selection	USN-3	As a user, I can select source and target languages.	2	High
Sprint 1	Text Translation	USN-4	As a user, I can enter text and get translated output instantly.	3	High
Sprint 2	Speech Translation	USN-5	As a user, I can speak input and receive translated text output.	3	Medium
Sprint 2	File Translation	USN-6	As a user, I can upload a document for translation.	3	Medium
Sprint 2	Translation History	USN-7	As a user, I can view previously translated texts.	2	Low
Sprint 3	AI Accuracy Optimization	USN-8	As a system, I improve translation accuracy using AI refinement techniques.	5	High

Project Tracker, Velocity & Burn down Chart (TransLingua)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint 1	10	7 Days	01 Mar 2025	07 Mar 2025	10	07 Mar 2025
Sprint 2	8	7 Days	08 Mar 2025	14 Mar 2025	8	14 Mar 2025
Sprint 3	7	7 Days	15 Mar 2025	21 Mar 2025	7	21 Mar 2025
Sprint 4	4	5 Days	22 Mar 2025	26 Mar 2025	4	26 Mar 2025

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test Scenarios & Results

Test Case ID	Scenario (What to Test)	Test Steps (How to Test)	Expected Result	Actual Result	Pass / Fail
FT-01	Validate text input fields (source text)	Enter valid text (English sentence). Enter empty input. Enter special characters only.	Valid text is accepted. Error message shown for empty or invalid input.	System validates correctly.	Pass
FT-02	Check source and target language selection	Select valid source and target languages. Leave language unselected.	Translation works when both languages are selected. Error shown if not selected.	Works as expected.	Pass
FT-03	Generate translated output	Enter text. Select languages. Click "Translate".	Correct translated text displayed.	Correct translated text displayed.	Pass
FT-04	Validate external translation API connectivity	Trigger translation request. Monitor API response.	API responds successfully.	API responds with status 200.	Pass
PT-01	Response Time Test	Use a timer to check content generation time.	Should be under 3 seconds.	Average 1.8 seconds.	Pass
PT-02	API Speed Test	Send multiple API calls at the same time.	API should not slow down.	Stable under moderate load.	Pass
PT-03	Large Text Processing Test	Translate large paragraph (1000+ words).	Successful translation without timeout.	Completed within acceptable time.	Pass

7. RESULTS

7.1 Output

Screenshots:

TransLingua - AI Powered Universal Translator

Translate text between any languages

 Enter text to translate:

 Enter target language (e.g., Hindi, Spanish, Japanese, Arabic...)

 Translate

TransLingua - AI Powered Universal Translator

Translate text between any languages

 Enter text to translate:

hello

 Enter target language (e.g., Hindi, Spanish, Japanese, Arabic...)

chinese

 Translate

 Translation Completed

Detected Language: English

Translation: 你好

8. ADVANTAGES & DISADVANTAGE

Advantages

1. Time-Saving

Provides instant multilingual translation without manual dictionary lookup.

Eliminates the need to use multiple translation tools.

Automatically detects source language and generates translation instantly.

2. Context-Aware Translation

Uses Google Gemini (LLM) for intelligent and natural translations.

Preserves sentence structure and meaning.

Produces more accurate results than simple word-to-word translators.

3. User-Friendly Interface

Built using Streamlit for easy interaction.

Simple input–output workflow.

Suitable for students, professionals, researchers, and travelers.

4. Multi-Language Support

Supports translation across multiple global languages.

Can dynamically handle different language pairs.

5. Scalability

Built on cloud-based AI APIs.

Can handle multiple users simultaneously.

Can be deployed on Streamlit Cloud, AWS, or other platforms.

6. Real-World Application

Helps students understand foreign study materials.

Assists businesses in global communication.

Supports travelers in foreign countries.

Useful for research collaboration across languages.

Disadvantages

1. Dependence on AI Accuracy

Translation quality depends on the AI model.

Ambiguous or unclear input may produce less accurate results.

2. Internet Dependency

Requires an active internet connection to access Gemini API.

Cannot function fully in offline mode.

3. API Usage Limits

Performance depends on API rate limits and subscription plans.

High usage may incur additional costs.

4. Limited Context in Some Cases

Very technical or domain-specific text may require refinement.

Idioms or cultural phrases may not always translate perfectly.

5. Data Privacy Considerations

User inputs are processed by external AI services.

Proper API key handling and security measures are necessary.

9. CONCLUSION

TransLingua: AI-Powered Multi-Language Translator successfully demonstrates the practical implementation of Generative AI (Google Gemini) integrated with a Streamlit web interface to solve real-world language communication challenges.

The system simplifies multilingual communication by automatically detecting source language and generating context-aware translations instantly. It reduces communication barriers in academic, professional, and personal contexts.

The project highlights how Large Language Models (LLMs) can be used effectively in Natural Language Processing (NLP) applications to improve accessibility, enhance productivity, and support global connectivity.

Overall, TransLingua showcases the potential of AI-driven language solutions to make cross-language communication faster, smarter, and more efficient.

10. FUTURE SCOPE

The system can be further enhanced to improve functionality, usability, and real-world applicability:

1. Speech-to-Text Translation

Enable users to speak input text.

Convert speech into translated text using AI.

2. Document Translation

Allow upload of PDF, Word, or text files for bulk translation.

3. Translation History

Store past translations for user reference.

Provide searchable translation history.

4. Multi-Language UI Support

Interface available in multiple languages.

Improve accessibility for global users.

5. Mobile Application Development

Develop Android/iOS app for on-the-go translation.

6. Real-Time Chat Translation

Enable live conversation translation between users.

Useful for international meetings and collaboration.

7. Integration with External Applications

API-based integration with websites, chat apps, or enterprise systems.

SaaS-based enterprise translation solutions.

8. AI Learning from User Feedback

Incorporate feedback mechanism.

11. APPENDIX

11.1 Source Code

```
import streamlit as st
from google import genai

# Initialize Gemini Client
client = genai.Client(api_key="Api Key")

st.set_page_config(page_title="TransLingua - AI Translator", page_icon="🌐")

st.title("🌐 Translingua - AI Powered Universal Translator")
st.write("Translate text between any languages")

# User Input
text = st.text_area("📝 Enter text to translate:")

target_language = st.text_input("🌐 Enter target language (e.g., Hindi, Spanish, Japanese, Arab:

if st.button("🚀 Translate"):
    if text.strip() and target_language.strip():
        try:
            prompt = f"""
            1. Detect the language of the following text.
            2. Translate it into {target_language}.
            3. Return the response in this clean format:

            Detected Language:
            <language name>
            """
```

```
            Translation:
            <translated text>

            Text:
            {text}
            """

            response = client.models.generate_content(
                model="gemini-2.5-flash", # use your working model name
                contents=prompt
            )

            # Clean UI Output
            st.success("✅ Translation Completed")
            st.markdown(response.text)

        except Exception as e:
            st.error(f"Error: {str(e)}")
    else:
        st.warning("Please enter both text and target language.")
```

11.2 Github Link

<https://github.com/harshithakondapaneni/TransLingua-AI-Powered-Multi-Language-Translator>

11.3 Project Demo Link

https://drive.google.com/file/d/1KJROTMAYhQ1LbWxNkFU-_hOTjXn6edKI/view?usp=drivesdk